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NPTEL ONLINE CERTIFICATION COURSE

SUSTAINABLE TRANSPORTATION SYSTEMS

Lecture 8: Impacts of Transportation Systems-III

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  - Direct Impacts
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- Potential Impacts during Life cycle of a Road/ Railway track
- Mitigation measures



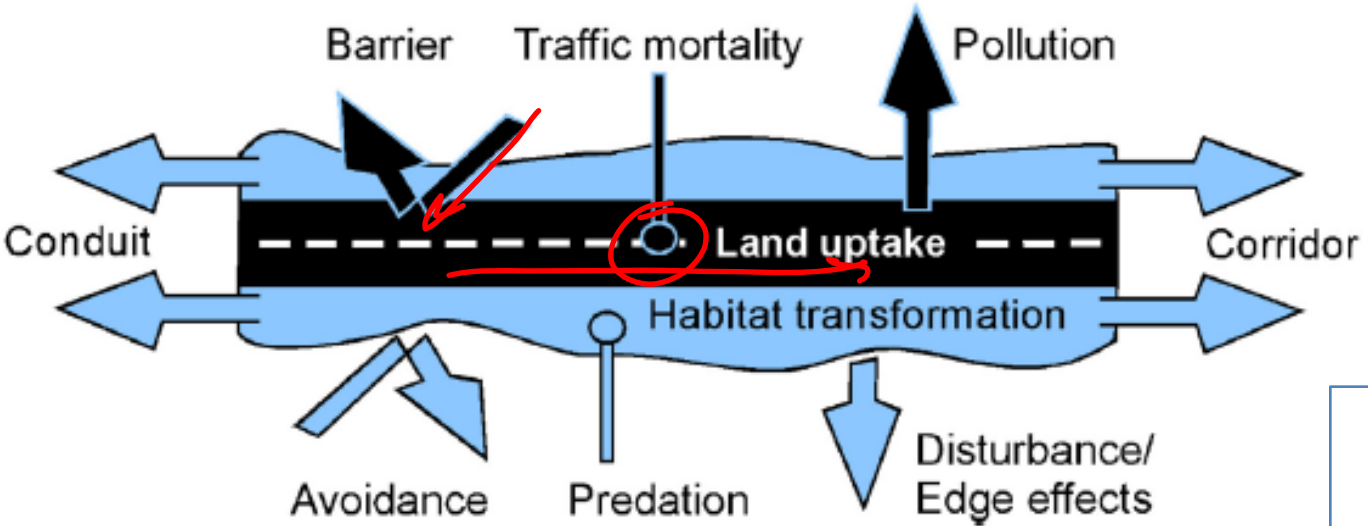
Impacts on Flora and Fauna

- Direct Impacts:
  - Loss of Wildlife Habitat
  - Habitat Fragmentation
  - Fauna Traffic Mortality
  - Corridor Restrictions (The Barrier effect)
  - Disturbance and Pollution
  - Traffic Noise
  - Visual Impacts
  - Interruption of biogeochemical cycle
- Indirect Impacts
  - Accessibility
  - Ecological Disequilibrium
  - Contamination of the biota
  - Forest fires



Impacts on Flora and Fauna

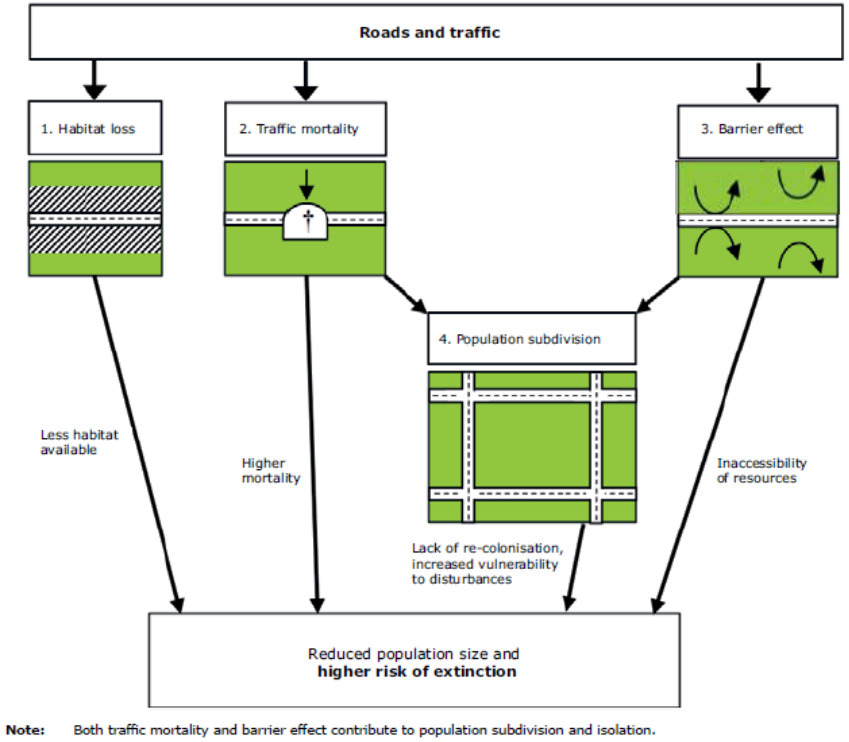
Direct Impacts



Schematic representation of the impacts of Transport Infrastructure on ecology

Source: (Andreas Seiler, 2003)

Impacts on Flora and Fauna



Note: Both traffic mortality and barrier effect contribute to population subdivision and isolation.

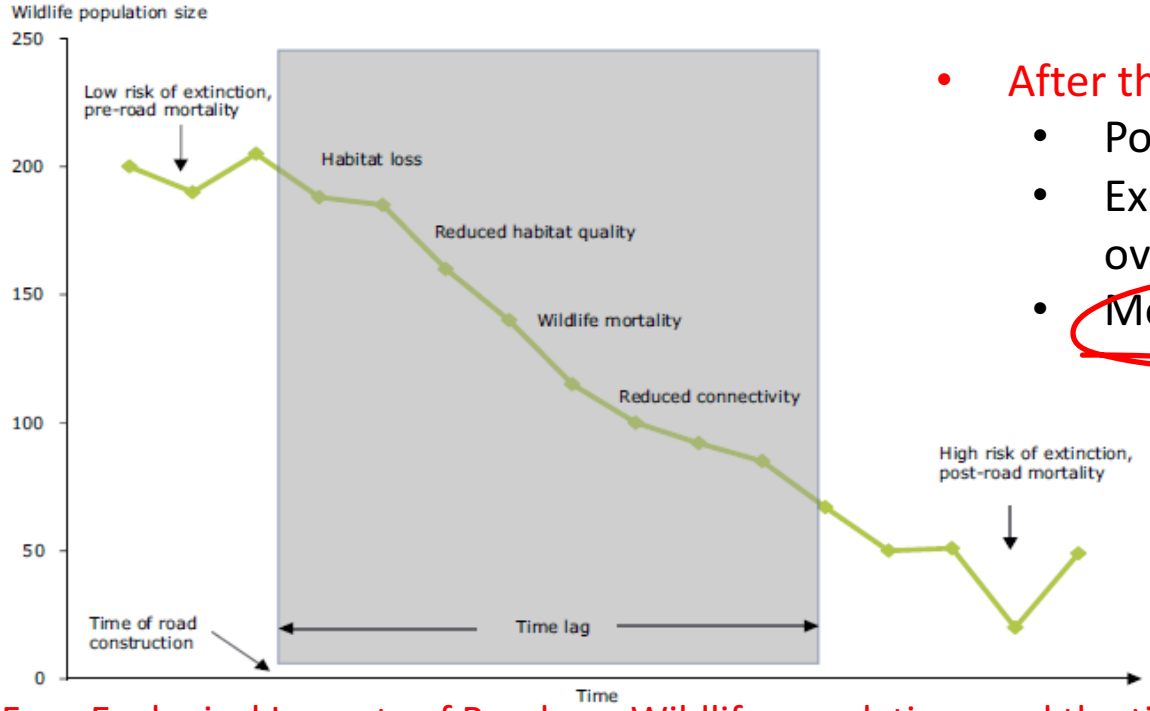
Source: (EEA, 2011)

Four Main Effects of Transportation Infrastructure on Wildlife

1. Habitat Loss
2. Traffic Mortality
3. Population Subdivision
4. Barrier effect

These effects reduces population size and increases risk of extinction

The Cumulative effect of Impacts on Wildlife



- After the time lag (in order of decades),
  - Population size is smaller
  - Exhibits greater relative fluctuations over time
  - More vulnerable

Four Ecological Impacts of Roads on Wildlife populations and the time lag for Cumulative effect

Source: (EEA, 2011)

Impacts on Flora and Fauna

Direct Impacts: Loss of Wildlife Habitat

- The natural habitat gets significantly altered by transport Infrastructure.
- May seem negligible at larger scales, but quite serious at local scale, depending on
  - landscape location of the infrastructure and
  - the affected habitat and species
- The transport infrastructure on hilly areas have greater consequences compared to other landscapes



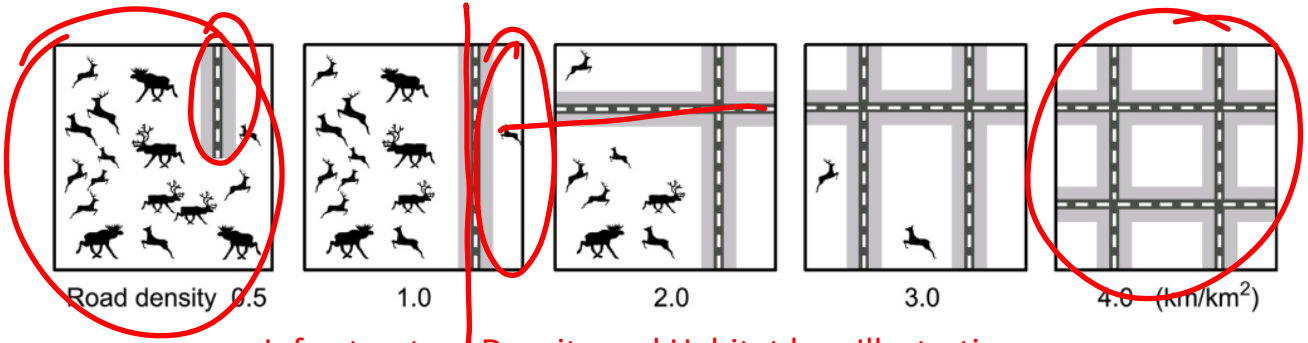
Source: (TRANSGREEN, 2019)

Impact on flora and fauna

Direct Impacts: Loss of Wildlife Habitat

- Infrastructure causes a loss and degradation of habitat due to disturbance effects (grey corridors) and isolation.

- With increasing infrastructure density, areas of undisturbed habitat (white) are reduced in size and become inaccessible.



Infrastructure Density and Habitat loss Illustration

Source: (Andreas Seiler, 2003)

Impact on flora and fauna

Direct Impacts: Loss of Wildlife Habitat

- Remnant fragments of suitable habitat may eventually become too small and isolated to prevent local populations from going extinct.

- The critical threshold in road density is species specific, but will also depend on landscape and infrastructure characteristics.






Source: (Andreas Seiler, 2003)

Impact on flora and fauna

Direct Impacts: Habitat fragmentation (The Barrier effect)



- Consequence of transport infrastructures causing impermeability of roads or railways for animals.

- Roads with high density traffic conditions and high-speed railways makes it impossible for most of the species to pass freely.

- Limits their ability to search for food, shelter, and mating characteristics.

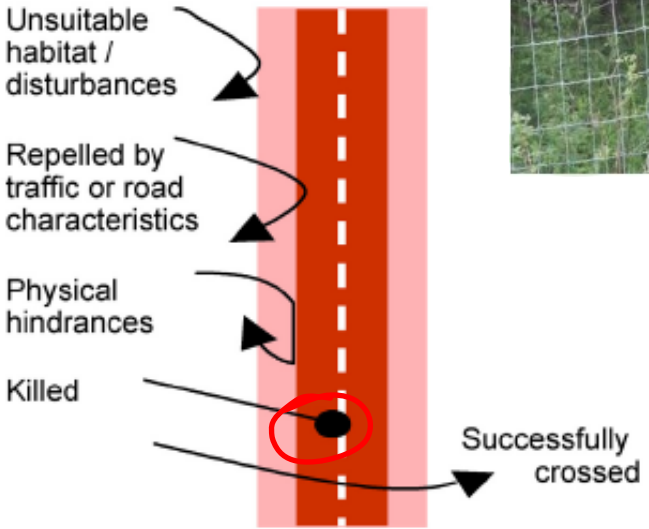
Source: (TRANSGREEN, 2019)

Impact on flora and fauna

Direct Impacts: Habitat fragmentation (The Barrier effect)



- Physical barriers includes
  - Completely fenced roads and railways
  - Unsuitable surfaces or crossing objects such as road verges

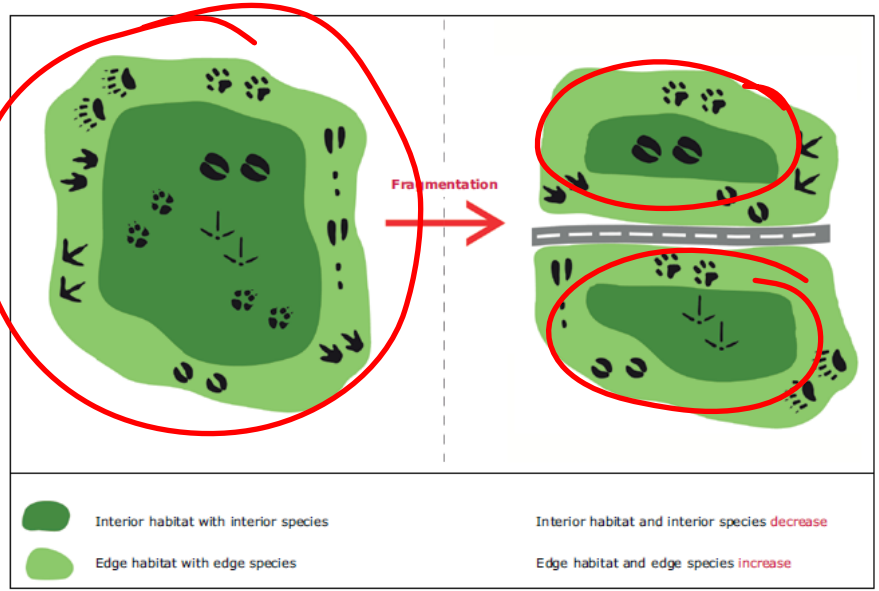


Source: (TRANSGREEN, 2019; Andreas Seiler, 2003)

Impact on flora and fauna

Direct Impacts: Habitat fragmentation (The Barrier effect)

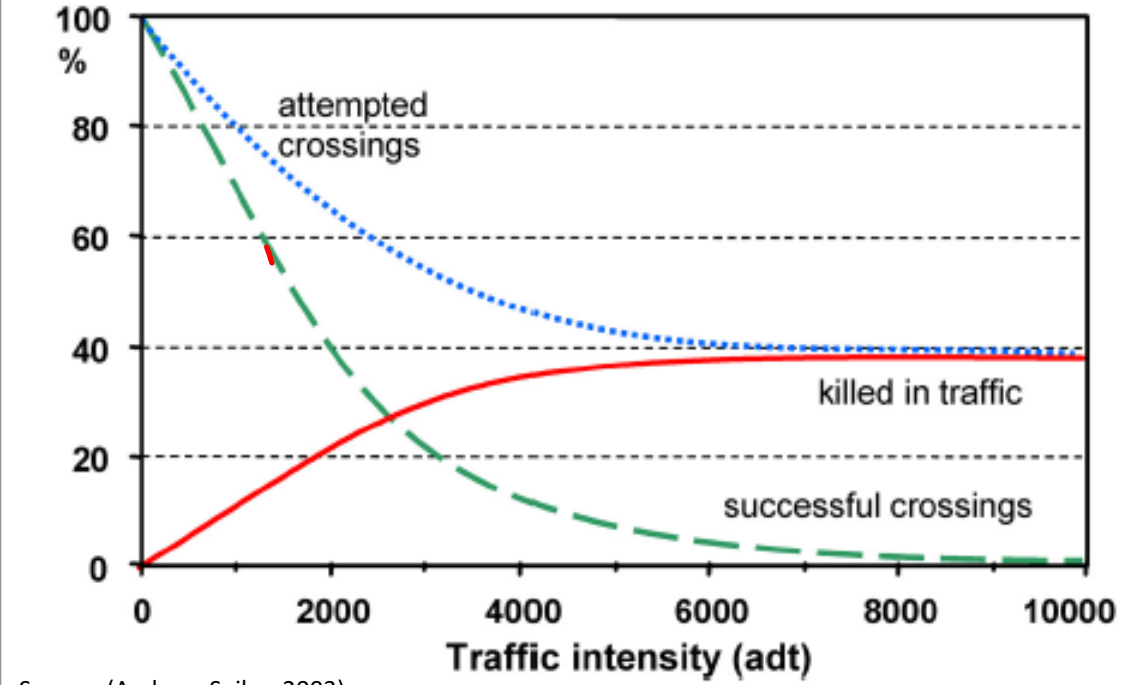
- When a road cuts through an ecosystem, the sum of the two parts created by the cut is less than the value of the initial whole.
- By slicing through habitat, roads affect an ecosystem's stability and health.
- Roads tend to fragment an area into weaker ecological sub-units, thus making the whole more vulnerable to invasions and degradation.



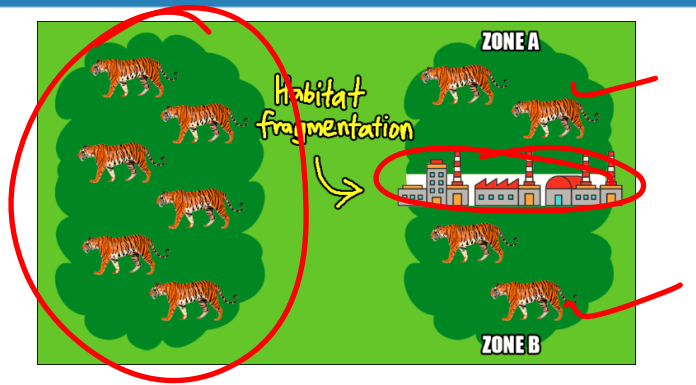
Source: (EEA, 2011)

Impact on flora and fauna

Theoretical Illustration of The Barrier effect



Source: (Andreas Seiler, 2003)



Increasing traffic increases road accidents linearly, until noise and vehicle movements repel animals from crossings.

Impact on flora and fauna

Direct Impacts: Fauna Traffic Mortality

- Most evident and well known negative impact of Transport Infrastructure.

- Mostly affected species includes:
  - Rare species that move long distances and are forced to use transport infrastructures. Ex. Otter, large carnivores such as tigers.
  - Seasonal migratory species such as amphibians (e.g., frogs, toads)

Source: (Andreas Seiler, 2003)



NH 67 (Gundlupet-Ooty Road)

Source: (http://www.conservationindia.org/)

Impact on flora and fauna

Direct Impacts: Fauna Traffic Mortality

- Concentration of fauna causalities on roads and railways depends on environmental factors such as:
  - Temperature
  - Precipitation or time of day
  - Ecological factors such as breeding dispersal, food supply, seasonal migrations, age and sex of animals etc.
  - Location, landscape of infrastructure, specifications of roads and traffic density, crop rotation etc.

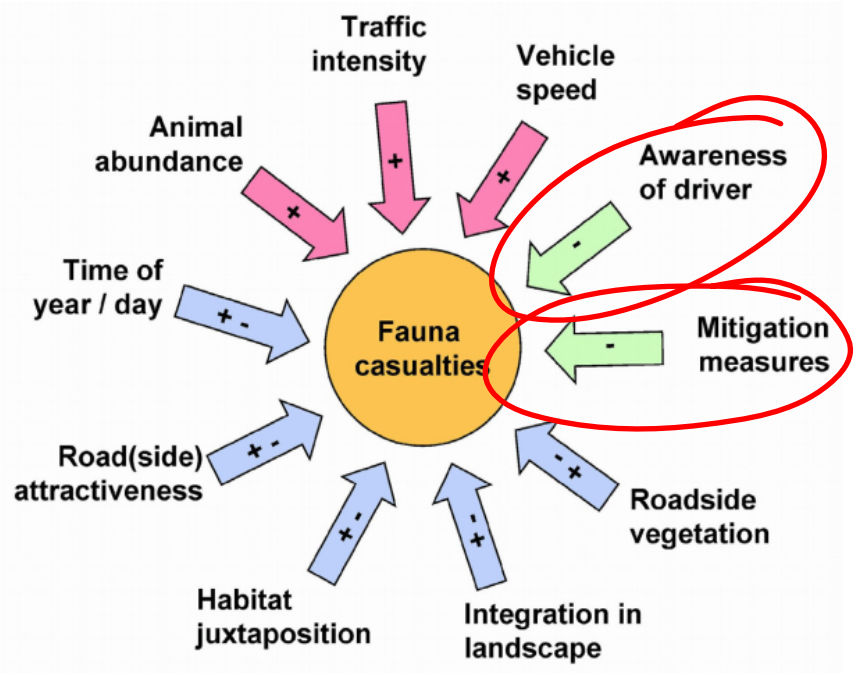


The National Highway 212 connecting Mysore in Karnataka and Kozhikode in Kerala

Source: (Andreas Seiler, 2003)

Impacts on Flora and Fauna

Direct Impacts: Fauna Traffic Mortality



Source: (Andreas Seiler, 2003)

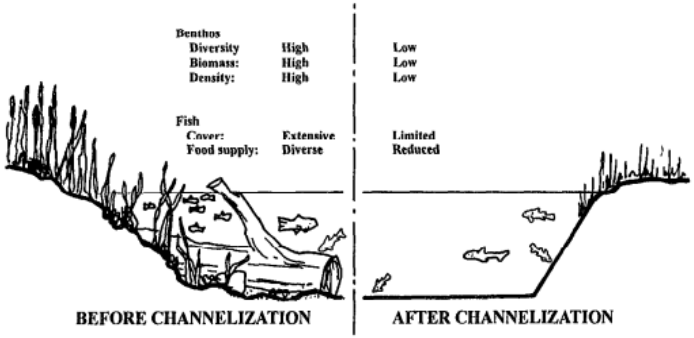
Factors Influencing the number of Wide life Traffic Accidents positively and negatively

Impact on flora and fauna

Direct Impacts: Aquatic habitat damage:

Effect of habitat fragmentation

- Transport development has very serious effects on aquatic ecosystems.



Source: Simpson et al, 1982

- Erosion from poorly constructed and rehabilitated sites can lead to downstream siltation, ruining spawning beds for fish.

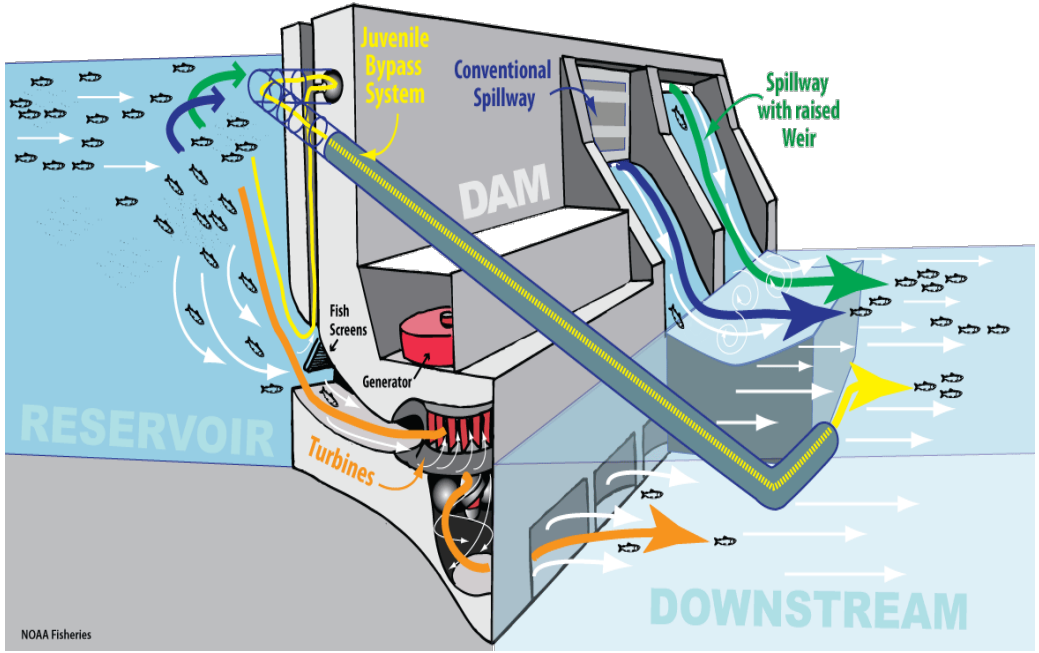
- Constricted channel flows at water crossings can make the current too fast for some species.

Source: (Andreas Seiler, 2003)

Impact on flora and fauna

Direct Impacts: Aquatic habitat damage:

Effect of habitat fragmentation



Source: (Andreas Seiler, 2003)

- Alterations of flood cycles, tidal flows, and water levels can upset trophic dynamics by affecting the life cycle of plankton, and have corresponding effects on the rest of the food chain.

Impact on flora and fauna

Direct Impacts: Aquatic habitat damage:

Effect of habitat fragmentation

Behavioral barriers

- Occur mostly in larger species
- Avoidance patters in animals such as not using areas near roads or railways constructed, avoid crossing large open spaces etc.



Source: (TRANSGREEN, 2019)

Impact on flora and fauna

Direct Impacts: Corridor restrictions

- Most animal species tend to follow established patterns in their daily and seasonal movements.

- When a road intersects or blocks a wildlife corridor, the result is either
  - A cessation of use of the corridor because animals are reluctant to cross the road,
  - An increase in mortality because of collisions with vehicles
  - A delay in migration which may result in the weakening or disappearance of an entire generation of the population.



Source: (Andreas Seiler, 2003)

Impact on flora and fauna

Direct Impacts: Corridor restrictions

- On busy roads, the death rate for the local amphibian or other slow-moving animal populations can be as high as one in ten.



Wild-life corridor- Crossings for animals

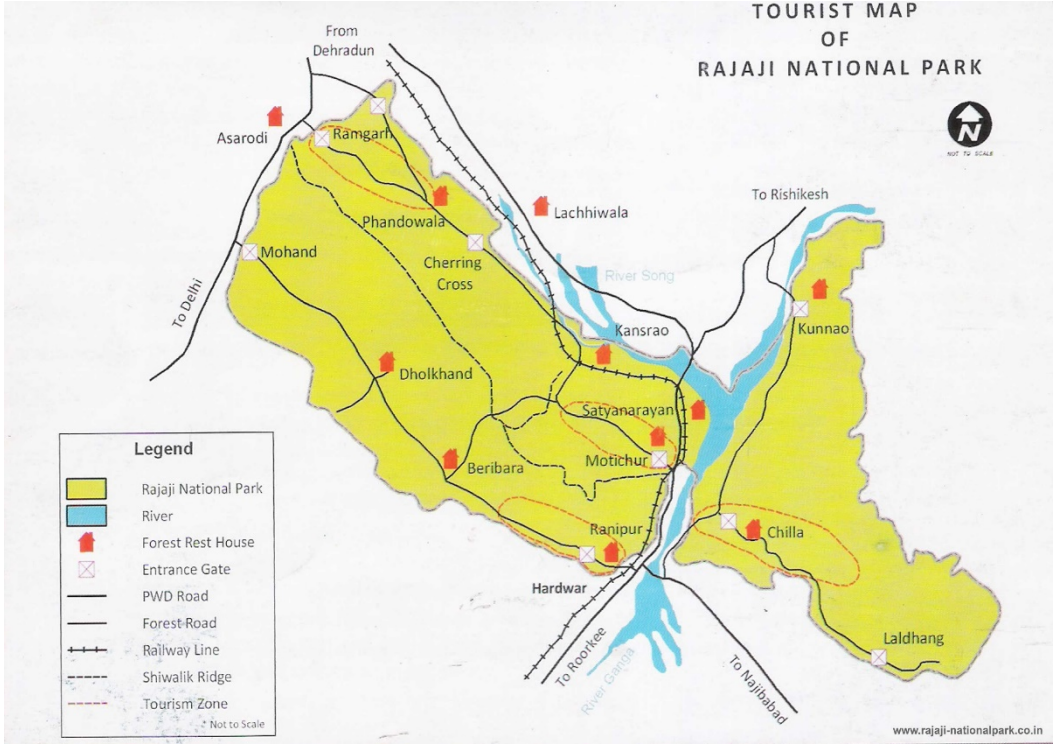


- The white markings on the overpass are track beds to record animal tracks crossing the overpass

Source: (EEA, 2011)

Impact on flora and fauna

Direct Impacts: Corridor restrictions



Source: (https://rajaji-nationalpark.co.in/images/map_rajaji.jpg)



Transport routes running along Rajaji national park, Uttarakhand, India

Impact on flora and fauna

Direct Impacts: Disturbance and Pollution

- Mountain environment is more sensitive to disturbances than landscape characteristics.
- Some disturbances includes:
  - Hydrological changes such as cuttings and embankments which increases soil erosion, changes in water regime etc.
  - Chemical pollution
    - Polluting oxides, hydrocarbons, particulates and heavy metals released from motor exhausts.
    - De-icing salt in winters causes pollution from sodium and chloride.
    - Contamination from herbicides during summers
    - Gasoline and oil leak during accidents



Source: (TRANSGREEN, 2019)

Impacts from Water Transport

Olivia Oil Spill 2011, Tristan da Cunha



The 40,170 GT bulk carrier OLIVIA grounded on the Nightingale Island, part of Tristan da Cunha, South Atlantic.

- Leaked 1400 tonnes of Intermediate Fuel and 70 tonnes of Marine diesel
- Oiled approx. 3700 Endangered Northern Rockhopper penguins

Source: (ITOPF, https://www.itopf.org/in-action/case-studies/case-study/flinterstar-belgium-2015/)

Impact on flora and fauna

Direct Impacts: Traffic Noise



- Traffic Noise is one of the major factors polluting natural environments.

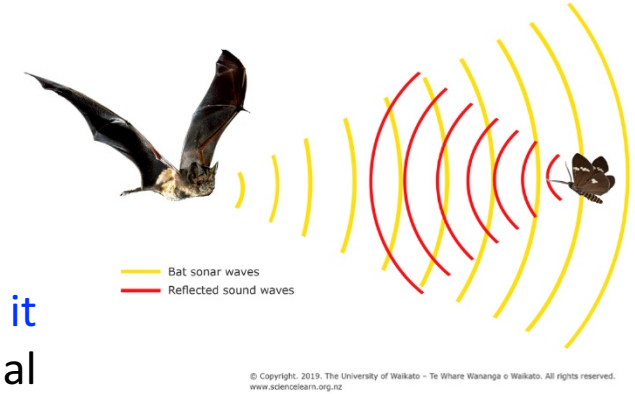
- Noise rarely has an immediate physiological effect on humans, however long exposure can induce psychological stress and eventually lead to physiological disorder.

- Most wild animal species interpret traffic noise as an indicator of the presence of humans and consequently avoid noisy areas.
  - Ex. Wild reindeer, Elk, Caribou and Brown bear avoid habitats near roads or utilise these areas less frequently.

Source: (Andreas Seiler, 2003)

Impact on flora and fauna

Direct Impacts: Traffic Noise



- Birds are very sensitive to traffic noise.
- Density of birds are minimum near trafficked roads, as it directly interferes with their vocal communication, territorial behaviour and mating success.
- Environmental factors such as the structure of verge vegetation, the type of adjacent habitat, and the relief of the landscape will alter the amplitude of the noise impact thereby influencing both noise spread and species density.

Source: (Andreas Seiler, 2003)

Impact on flora and fauna

Direct Impacts: Traffic Noise



- The species most vulnerable to disturbance and development impacts includes:
  - Large species
  - Long-lived species
  - Species with relatively low reproductive rates
  - Habitat specialists
  - Species living in open (wetlands) rather than closed (forest) habitats
  - Rare species
  - Species whose populations are concentrated in a few areas

Source: (Andreas Seiler, 2003)

Impact on flora and fauna

Direct Impacts: Visual Impacts



- The effects of traffic also include Visual disturbance
  - Ex. Artificial lights or vehicle movement in forest roads.

- Artificial lighting has a conflicting effect on different species of fauna and flora:
  - It can act as a deterrent to deer and a readily accessible insect food supply to bats, but at the same time it can disrupt growth regulation in plants, breeding and behaviour patterns in birds, bats, nocturnal frogs and moth populations.

Source: (Andreas Seiler, 2003)

Impact on flora and fauna

Direct Impacts: Visual Impacts



- White (mercury vapour) street lamps are especially attractive to insects, and therefore also to some aerial-hawking bat species.

- This increases the exposure of bats to traffic and may entail increased mortality due to collisions with vehicles.

- The movement of vehicles (in combination with noise) can alter behaviour and induce stress reactions in wildlife.
  - Ex. Heavy trucks and high-speed trains produce intensive discontinuous noise, vibration and visual disturbance which has the effect of frightening many mammals and birds.

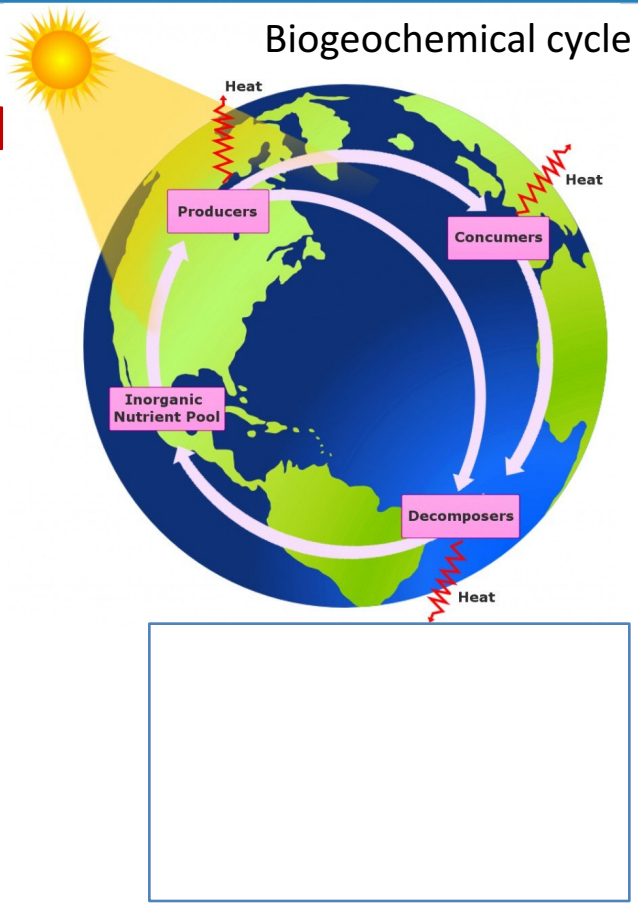
Source: (Andreas Seiler, 2003)

Impact on flora and fauna

Direct Impacts: Interruption of biogeochemical cycle

- The flow of nutrients and materials is a major determinant in ecosystem structure and function.

- Road development can easily disrupt it through
  - Alteration of flows of surface and groundwater
  - Removal of biomass
  - Relocation of topsoil.

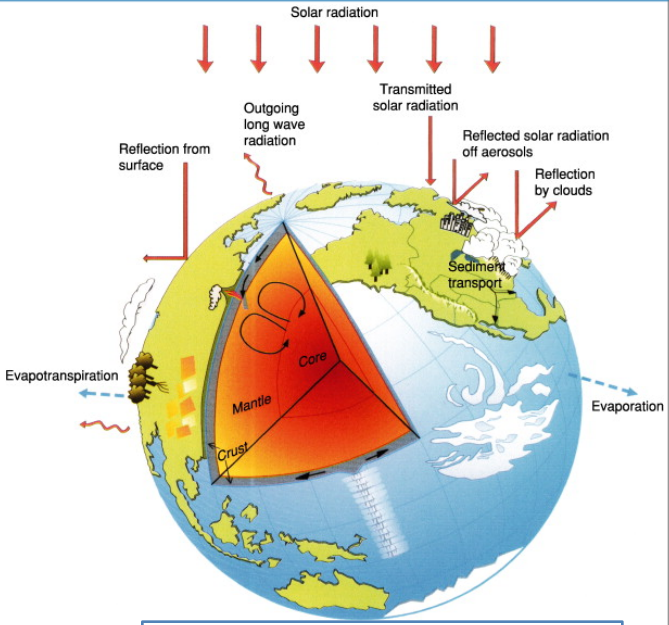


Source: (World Bank, 1997)

Impact on flora and fauna

Direct Impacts: Interruption of biogeochemical cycle

- Human activity can be a major source of nutrients
  - may increase turbidity and BOD.
- Potential impacts can only be roughly estimated based on:
  - Nature of the alteration has been established
  - Data on soil erodibility
  - Soil fertility
  - Anticipated human activity
  - Understanding of the nutrient regime and energy flow of the affected ecosystem



Source: (World Bank, 1997)

Impact on flora and fauna

Indirect Impacts: Accessibility

- Roads increase contact between humans and the natural environment, and leads to ecosystem modification.
- Vulnerable to human activities such as recreation, forest and mineral exploitation to colonization and urbanization.
- Upgrading of existing roads increases the number of people having accessibility and increases the likelihood of impacts.



Source: (World Bank, 1997)

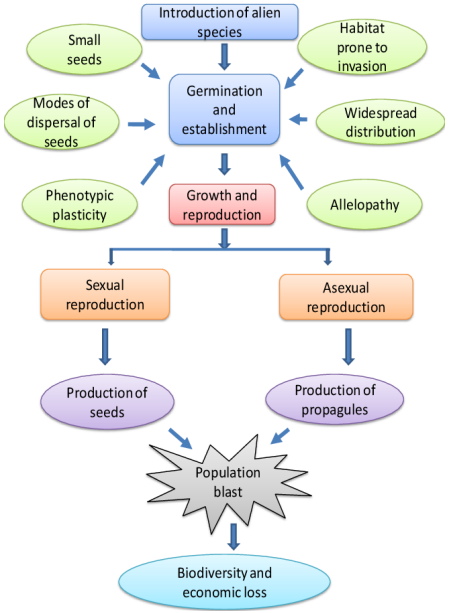
Impact on flora and fauna

Indirect Impacts: Ecological disequilibrium

- The importation of new plant and animal species along the right-of-way can upset the dynamic balance which exists in ecosystems.

- Native species face competition for resources from new arrivals, and predator-prey relationships can be altered, often to the detriment of the native species. Non-native species can gain a competitive advantage because of a lack of natural controls and become dominant.

- The result is usually a simplified ecosystem which is more vulnerable to further impacts.



Source: (World Bank, 1997)

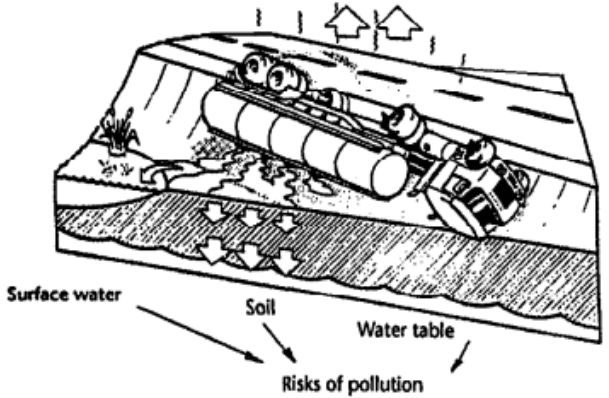
Impact on flora and fauna

Indirect Impacts: Contamination of the biota

- The presence of motor vehicles introduces the potential for contamination of the soil, air, and water adjacent to the road and in the case of surface water, well beyond the immediate surroundings.

- Chronic contamination can become a serious problem for animal species, especially those at the top of the food chain, because of bioaccumulation of pollutant.

Source: (World Bank, 1997)



Risk of contamination from accidents.



Contamination of water and air from accidents

Impact on flora and fauna

Indirect Impacts: Forest Fires

- Increases in human activity are associated with more frequent incidents of fires, which can have sudden, severe, and wide ranging impacts.



Forest fire in Dehradun, Uttarakhand, India

(Source: https://thelogicalindian.com/news/national-green-tribunal-bans-chopping-of-trees-in-uttarakhand-for-construction-of-roads/)





Fires due to Road accidents

Source: (World Bank, 1997)

Impacts on Crop yield



- Rising temperatures and increased emissions of long lived GHGs have significant negative impacts on Crop yields.

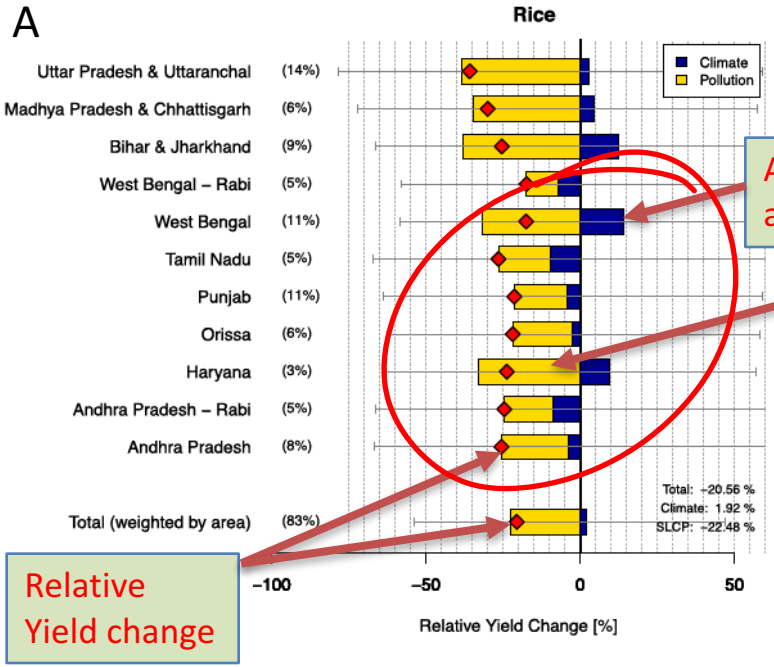
- Short lived Black carbon and ozone impacts temperature, precipitation patterns, radiations in an area and can be toxic to plants.

- Ex. Wheat yield in India in 2010 were 36% lower compared to the 1980s with some densely populated states experiencing 50% relative yield losses.

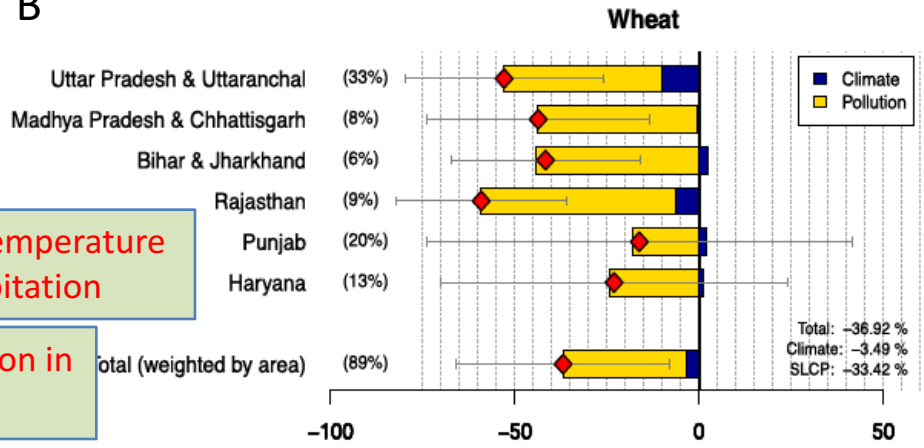
Source: (Jennifer Burney and V. Ramanathan, 2013)

Impacts on Crop yield

A



B



Average temperature and precipitation

Reduction in Yield

Relative Yield change

The Relative yield change calculated as 2006-2010 average for (A) Rice and (B) Wheat

- 90% of the reduction in Wheat yield can be attributed to Short lived Climate pollutants

Source: (Jennifer Burney and V. Ramanathan, 2013)

Potential Impacts during the Life cycle stages of a Road and Railway

	Life cycle stage	Potential problems
a	Planning	Quality of Routing Availability of biodiversity / ecological data
b	Construction	Destruction of natural habitats Creation of new artificial habitats Effect on groundwater and surface water Mortality of animals at constructive sites Noise and other emissions and contamination
c	Operation	Barrier effect of roads/ railways Mortality of animals, Human causalities and damages Gaseous, liquid and solid emissions from transport Noise pollution Road maintenance
d	Removal	Processing and elimination of waste materials

Source: (TRANSGREEN, 2019)

Mitigation Measures

Fencing



Culvert with small mesh sized fencing

Source: (Clara Grilo et. al, 2010)

Mitigation Measures

Crossing Structures



Small Under passes (Metal or concrete pipe culverts for smaller animals)



Intermediate sized underpasses (For medium sized animals such as foxes, black bear etc.)

Source: (Clara Grilo et. al, 2010)

Mitigation Measures

Fencing



Underpass with 1.5 m fencing to prevent animal entry on roads

Source: (Clara Grilo et. al, 2010)

Mitigation Measures

Crossing Structures





Larger Under passes

(For Large carnivores and other animals)

Source: (Clara Grilo et. al, 2010)

Mitigation Measures

Crossing Structures



Source: (Clara Grilo et. al, 2010)

Extensive bridges

- For large carnivores like grizzly bears and wolves

Mitigation Measures

Crossing Structures



Source: (Clara Grilo et. al, 2010)

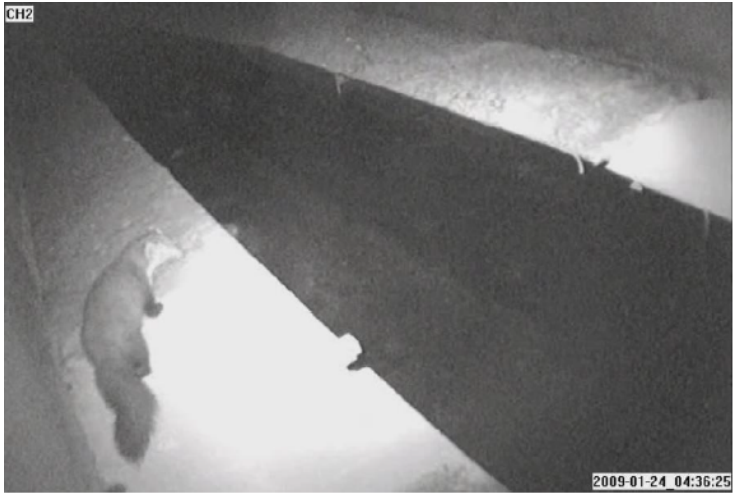
Wildlife overpasses

- Allows full sunlight and precipitation
- Full vegetation growth similar to forests

Mitigation Measures

Dry ledges





Stone marten using a dry ledge to cross under a road in southern Portugal

Box culvert with dry ledges on sides for passage of smaller animals

Source: (Clara Grilo et. al, 2010)

Mitigation Measures

Escape opportunities from Right of way



Earthen ramps for animals to escape from accidents and for visibility of drivers

Source: (Clara Grilo et. al, 2010)

Mitigation Measures

Amphibian tunnel

Top graded amphibian tunnel

- Open top allows light and air flow



Source: (Clara Grilo et. al, 2010)

Mitigation Measures

Paynes Prairie wall



Paynes Prairie wall to prevent Turtles from accessing the roads in Portland

Source: (Clara Grilo et. al, 2010)

Conclusions

- The Transportation activities can significantly harm the Flora and Fauna of an ecosystem.

- Activities such as construction activities and traffic through forests, and the light and noise disturbances can have huge impact on animals and plants in the area, disturbing their daily movements, causing fragmentation of their habitats, accidents and reducing their populations.

- Careful planning is the only solution for reducing the transport-related impacts, preserving the local flora and fauna.

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Thank You